PAPER<i>274: HI 21-cm Line as UQFF Galactic Buoyancy Resonance Frequency — ω</i>HI Bridges Atomic Hyperfine Physics to Galaxy-Scale Dynamics

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Session: 75 — Andromeda UQFF 2.0 Analysis

PAPER*274: HI 21-cm Line as UQFF Galactic Buoyancy Resonance Frequency* — ωHI Bridges Atomic Hyperfine Physics to Galaxy-Scale Dynamics

Authors: Daniel T. Murphy **Date:** March 2026 **UQFF Module:** ANDROMEDA*UQFF*MODULE.cpp (M31 Master Module, Session 75) **Session:** 75 — Andromeda UQFF 2.0 Analysis **Keywords:** 21-cm HI line, hydrogen spin-flip, hyperfine transition, UQFF resonance, galactic buoyancy frequency, omega*HI, 1.42 GHz, nu*HI, atomic-to-galactic bridge

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Abstract

The neutral hydrogen 21-cm spin-flip transition at ν_{HI} *= 1.42040575 GHz is one of the most precisely known frequencies in all of physics and is universally used as a velocity tracer in radio astronomy. In this paper, we demonstrate that this frequency appears naturally in the UQFF framework as the* ***galactic buoyancy resonance frequency****: when the resonant oscillatory term of the master UQFF gravity equation is parameterized with* $\omega = \omega_{HI} = 2\pi \times 1.42040575 \times 10$■ rad/s, it produces a resonant gravitational force $F_{res}(t) = A_{res} \times \cos(\omega_{HI} \times t) \times \exp(-t/\tau_{gal})$ that is simultaneously consistent with both the atomic hyperfine energy splitting in hydrogen ($E_{HF} = h\nu_{HI} = 9.411\times10^{-2}$■ J) and the large-scale buoyancy dynamics of galaxy-sized systems. We identify ω_HI as the **HI-UQFF Bridging Frequency**, constituting a new multi-scale coupling between quantum atomic physics and gravitational galaxy dynamics within the UQFF framework.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Introduction

The hydrogen 21-cm line arises from the hyperfine transition between the F=1 (parallel electron-proton spin) and F=0 (antiparallel) states of the hydrogen ground state. Its frequency is:

```
\nu_\text{HI} = 1.42040575177 \times 10^9\ \text{Hz}
```

```
\omega_\text{HI} = 2\pi \times \nu_\text{HI} = 8.92819 \times 10^9\ \text{rad/s}
```

This transition has zero classical analogue and arises purely from quantum electrodynamic effects (magnetic dipole coupling between electron and proton magnetic moments). It is measured to 12 significant figures and is used as a cosmic standard.

In UQFF, the galactic resonance term appears as:

```
F_\text{res}(t) = A_\text{res} \times \cos(\omega_\text{osc} \times t) \times
e^{-t/\tau_\text{gal}}
```

The key question: what is the natural value of ω_{osc} for a galaxy like Andromeda?

Our finding: **ω*osc* = ωHI = 8.92819 × 10■ rad/s** is the physically motivated choice, linking the atomic spin-flip frequency to the galactic buoyancy oscillator.

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2. Mathematical Formulation

2.1 UQFF Resonant Term

```
\boxed{F_\text{res}(t) = A_\text{res} \times \cos(\omega_\text{HI} \times t) \times
e^{-t/\tau_\text{gal}}}
```

where:

$A_{res} = 1.0\times10^{-12}$ m/s^2 (galactic resonance amplitude)

$\omega_{HI} = 2\pi \times 1.42040575\times10$■ = 8.92819×10■ rad/s

τ_{gal} = 1 Gyr = 3.15576×10^{1}■ s

2.2 Parameter Values

Parameter	Symbol	Value	Units
HI frequency	ν_{HI}	1.42040575×10■	Hz
Angular frequency	ω_{HI}	8.92819×10■	rad/s
Galactic period	*THI* = $2\pi/\omega$HI	7.037×10^{-10}	s
Hyperfine energy	*EHF* = ■ωHI	9.411×10^{-2}■	J
Resonance amplitude	A_{res}	1.0×10^{-12}	m/s^2
Galactic decay time	τ_{gal}	3.156×10^{1}■	s (1 Gyr)

2.3 computeHz() Implementation

The `computeHz()` method in `AndromedaUQFFModule` returns ω_{HI} directly:

```
double computeHz() { return omega_HI; } // 8.9282e9 rad/s
```

This encodes the 21-cm frequency as the canonical galactic UQFF frequency output, accessible via the public interface for use in multi-module cross-validation.

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3. Physical Motivation for ω_{HI} as UQFF Frequency

3.1 Hydrogen's Universal Presence

Hydrogen constitutes ~74% of the cosmic baryonic mass fraction. In Andromeda, neutral HI gas is distributed throughout the disk and halo with a total HI mass of ~6.5×10■ M_{sun} (~3.6% of total mass). The 21-cm emission traces the rotation curve, spiral structure, and velocity dispersion of the galaxy.

In UQFF, the buoyancy dynamics of a system are driven by its dominant mass constituents. Since hydrogen is the dominant visible-matter component (and its atomic spin-flip frequency ω*HI governs the characteristic oscillation time of HI gas), using* ωHI as the galactic resonance frequency is physically motivated.

3.2 HI-UQFF Scale Bridging

The 21-cm transition bridges two vastly different physical scales:

Scale	Physical Domain	Role of ω_HI
Atomic (10^{-10} m)	Hydrogen ground state hyperfine structure	Photon emission frequency
Galactic (10^{21} m)	Galaxy disk rotation and HI distribution	UQFF buoyancy resonance

The ratio of these scales is ~10^{31}, yet a single frequency ω_HI appears in both. This is the **HI-UQFF Bridging Frequency** — a scale-invariant constant of hydrogen atomic physics that re-emerges at galactic scale in the UQFF buoyancy framework.

3.3 Connection to Radio Velocity Tracing

In radio astronomy, the observed frequency of 21-cm emission is Doppler-shifted by the gas velocity:

$$\nu_\text{obs} = \nu_\text{HI} \left(1 - \frac{v_r}{c}\right)$$

This is inverted to measure v_r *(radial velocity). In UQFF, the* $\cos(\omega_{HI} \times t)$ oscillation in F_{res} *encodes the same velocity information: the phase of the oscillation at time t maps to the dynamical state of the HI gas at that epoch. At t = 0 (initial epoch),* $F_{res} = A_{res}$ *(maximum positive buoyancy); at* $t = \pi/\omega_{HI} \approx 0.35$ ns, $F_{res} = -A_{res}$ (maximum negative buoyancy). Over the 1 Gyr timescale τ_gal, the amplitude decays to ~0 as HI gas depletes through star formation.

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4. Evaluation at Andromeda Parameters

4.1 F_res at t = 0

$$F_\text{res}(0) = A_\text{res} \times \cos(0) \times \exp(0) = 1.0 \times 10^{-12}\ \text{m/s}^2$$

4.2 Fres at t = τgal = 1 Gyr

$$F_\text{res}(\tau_\text{gal}) = 1.0 \times 10^{-12} \times \cos(\omega_\text{HI} \times 3.156 \times 10^{16}) \times e^{-1}$$

The cosine argument: $\omega_\text{HI} \times \tau_\text{gal} = 8.928 \times 10^9 \times 3.156 \times 10^{16} = 2.818 \times 10^{26}$ radians — the resonance oscillates extremely rapidly (10■ Hz × Gyr ≈ 10^2■ cycles) and its time-average is zero. The exp(−1) = 0.368 envelope governs the long-term envelope.

This means **the HI resonance term oscillates at sub-nanosecond timescales while its amplitude envelope decays over Gyr** — an extreme multi-scale temporal structure unique to using ω_HI in a galactic context.

4.3 HI-UQFF Bridging Constant

We define:

$$\Omega_\text{bridge} \equiv \frac{\omega_\text{HI}}{\omega_g} = \frac{8.928 \times 10^9}{7.3 \times 10^{-16}} = 1.223 \times 10^{25}$$

where ω_g *= 7.3×10^{-1}■ rad/s is the canonical UQFF gravitational buoyancy frequency. The ratio* Ω_{bridge} = 1.223×10^2■ encodes the scale separation between atomic quantum oscillations and gravitational galaxy dynamics.

```
\boxed{\Omega_\text{bridge} = \frac{\omega_\text{HI}}{\omega_g} = 1.223 \times 10^{25}}
```

---

5. Uniqueness of ω_HI in UQFF Context

Unlike phenomenological choices of ω*osc (which could be set to any value)*, ωHI is:

- **Observationally anchored** — measured to 12 significant figures
- **Cosmically universal** — same frequency everywhere in the universe (barring z)
- **Mass-traced** — HI gas traces the dominant baryonic mass distribution
- **Quantum-derived** — arises from first principles of atomic physics (no free parameter)

No other astrophysical frequency combines all four properties simultaneously at the galactic scale.

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6. Conclusions

We identify the neutral hydrogen 21-cm spin-flip frequency as the natural UQFF galactic buoyancy resonance frequency for hydrogen-dominated galaxy systems:

```
\boxed{\omega_\text{HI} = 2\pi \times 1.42040575 \times 10^9\ \text{rad/s} = 8.92819
\times 10^9\ \text{rad/s}}
```

```
\boxed{F_\text{res}(t) = A_\text{res} \cos(\omega_\text{HI} t)\, e^{-t/\tau_\text{gal}}}
```

The **HI-UQFF Bridging Constant** Ω_bridge = 1.223×10^{2}■ quantifies the scale separation bridged by this single frequency. This is the first explicit UQFF connection between atomic hyperfine physics and galaxy-scale gravitational buoyancy dynamics.

Derived from ANDROMEDAUQFFMODULE.cpp, UQFF 2.0, Session 75. Next: PAPER_275 (DM 80/20 shell partition).

UQFF computed: MUGE buoyancy ratio U*bi*/*F*U = [SSq]■?■r■/GM = 5.7e-1■5.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s■ at r_ISCO.
